

3rd Semester cse Branch

DATA STRUCTURE AND ALGORITHMS

VVI OBJECTIVE



LAKSHYA BATCH *FOR CSE*



Start Date: **18th May 2026**

4TH SEMESTER



Er. Nadim | Er. Asif | Er. Hammad

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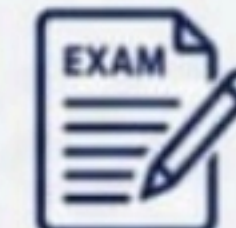
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Complete
Syllabus



Exam-Oriented
Approach



Regular
Doubt Solving

ENROLL NOW!



(1) Add Which asymptotic notation provides the tightest upper bound?

(A) $\Omega(n)$

✓ (B) $O(n)$ Big O

(C) $\theta(n)$

(D) $o(n)$ a heading

UP

↓

L.O

(2) The time complexity of finding an element in a sorted array of size n using Binary Search is:

(A) $O(n)$

(B) $O(n \log n)$

✓ (C) $O(\log n)$

(D) $O(n^2)$

(3) Space complexity of an algorithm is:

- ☒ **(A) Maximum memory used by the algorithm**
- (B) Average memory used**
- (C) Minimum memory used**
- (D) Memory used by the input only**

(4). Which data structure is used for implementing recursion?

- (A) Queue**
- ☒ **(B) Stack**
- (C) Linked List**
- (D) Array**

(5). A Deque (Double Ended Queue) allows:

(A) Insertion at both ends, deletion at one end

☒ (B) Insertion/Deletion at both ends

(C) Only one end access

(D) None of the above

(6) The postfix form of the infix expression $(A + B) * C$ is:

- ☒ (A) $AB+C^*$
- (B) $ABC+^*$
- (C) $+ABC$
- (D) $ABC+$

Postfix expression

(7) In a circular queue of size N, the 'Queue Full' condition is:

(A) $\text{front} == \text{rear}$

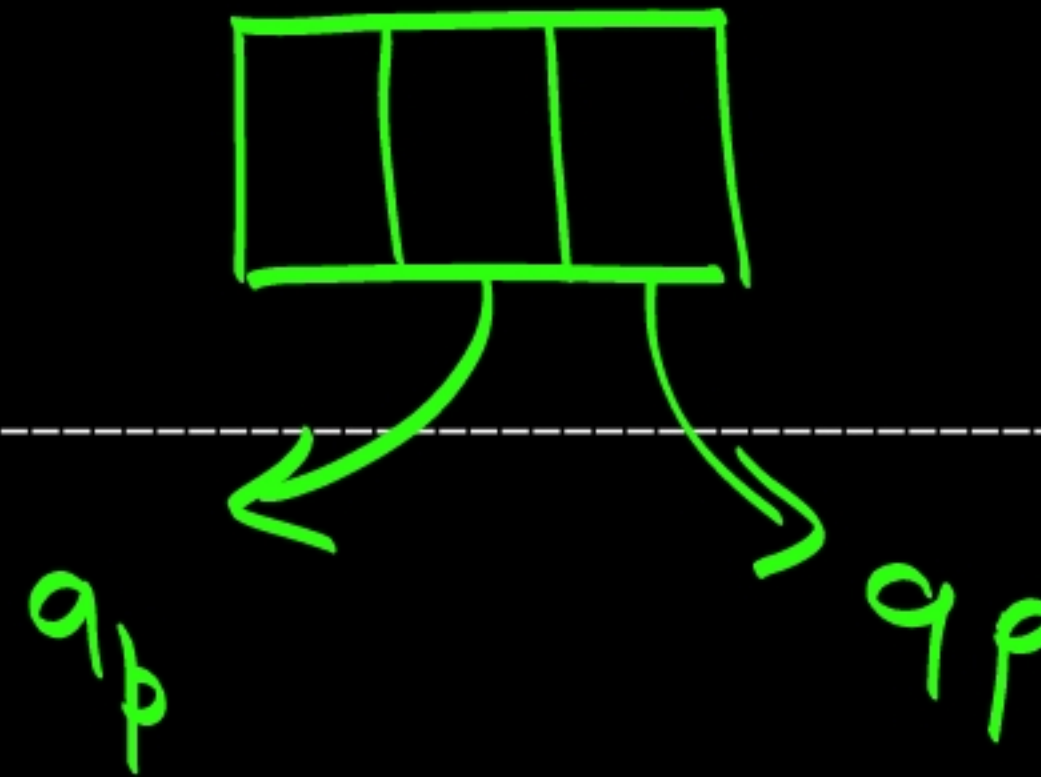
 **(B) $(\text{rear} + 1) \% N == \text{front}$**

(C) $\text{rear} + 1 == N$

(D) $\text{front} == (\text{rear} + 1) \% N$

(8) In a Doubly Linked List, the minimum number of pointers in a node is:

- (A) 1
- ☒ (B) 2
- (C) 3
- (D) 0



(9) Which linked list does not have a NULL pointer at the end?

(A) Singly Linked List

(B) Doubly Linked List

✓ (C) Circular Linked List

(D) Header Linked List

(10) Time complexity to insert a node at the beginning of a singly linked list is:

- (A) $O(n)$**
- (B) $O(\log n)$**
- ✓ (C) $O(1)$**
- (D) $O(n^2)$**

(11) An AVL tree is a self-balancing BST where the balance factor of any node is:

(A) Exactly 0

☒ **(B) -1, 0, or 1**

(C) Greater than 1

(D) Any integer

(12) The number of null pointers in a binary tree with n nodes is:

(A) n

(B) $n - 1$

✓ (C) $n + 1$

(D) $2n$

(13) Which traversal of a BST gives sorted output?

- (A) Preorder**
- (B) Postorder**
- ✓ (C) Inorder**
- (D) Level order**

(14) What is the maximum height of a binary tree with n nodes?

(A) $\log n$

(B) $n/2$

☒ (C) n

(D) $n - 1$

(15) A complete binary tree with n nodes has a height of:

(A) n

☒ (B) $\log_2(n + 1)$

(C) $n/2$

(D) n^2

(16) Which algorithm is used to find the Shortest Path from a single source?

(A) Kruskal' s

(B) Prim' s

☒ (C) Dijkstra' s

(D) Huffman Coding

(17) The number of edges in a Minimum Spanning Tree (MST) of a graph with V vertices is:

- (A) V**
- (B) $V + 1$**
- ☒ (C) $V - 1$**
- (D) $2V$**

(18) BFS (Breadth-First Search) uses which data structure?

(A) Stack

(B) Queue ✓

(C) Tree

(D) Heap

(19) DFS (Depth-First Search) uses which data structure?

- (A) Stack ✓**
- (B) Queue**
- (C) Linked List**
- (D) Hash Table**

(20) A graph with no cycles is called:

- (A) Bipartite graph**
- ☒ **(B) Acyclic graph**
- (C) Connected graph**
- (D) Complete graph**

(21) Worst-case time complexity of Quick Sort is:

(A) $O(n \log n)$

(B) $O(n)$

(C) $O(n^2)$ ✓✓

(D) $O(\log n)$

(22) Which sorting algorithm is stable and has $O(n \log n)$ worst-case complexity?

- (A) Quick Sort**
- (B) Selection Sort**
- ✓ (C) Merge Sort**
- (D) Bubble Sort**

(23) The best-case time complexity of Bubble Sort (optimized) is:

(A) $O(1)$

 **(B) $O(n)$**

(C) $O(n \log n)$

(D) $O(n^2)$

(24) Heap Sort is based on which data structure?

- (A) Balanced BST**
- (B) Stack**
- (C) Complete Binary Tree ✓**
- (D) Linked List**

(25) What is a collision in Hashing?

- (A) Two keys have different hash values**
- (B) Two keys have the same hash value ✓**
- (C) Hash table is empty**
- (D) Hashing function is complex**

(26) Which sorting algorithm uses the Divide and Conquer strategy?

(A) Insertion Sort

(B) Selection Sort

(C) Merge Sort ✓

(D) Heap Sort

(27) Linear probing is a technique for:

- (A) Overflow handling**
- (B) Collision resolution ✓✓**
- (C) Searching**
- (D) Sorting**

(28) In Hashing, Chaining is a technique of:

(A) Open Addressing

(B) Closed Hashing

(C) Open Hashing ✓

(D) None of the above

(29) A B-Tree of order m can have at most how many children?

(A) $m - 1$

 (B) m

(C) $m/2$

(D) $2m$

(30) What is the complexity of inserting an element into a Max Heap?

(A) $O(1)$

(B) $O(n)$

(C) $O(\log n)$ ✓✓

(D) $O(n \log n)$

(31) Which of the following is NOT a linear data structure?

(A) Array

(B) Stack

(C) Graph ✓

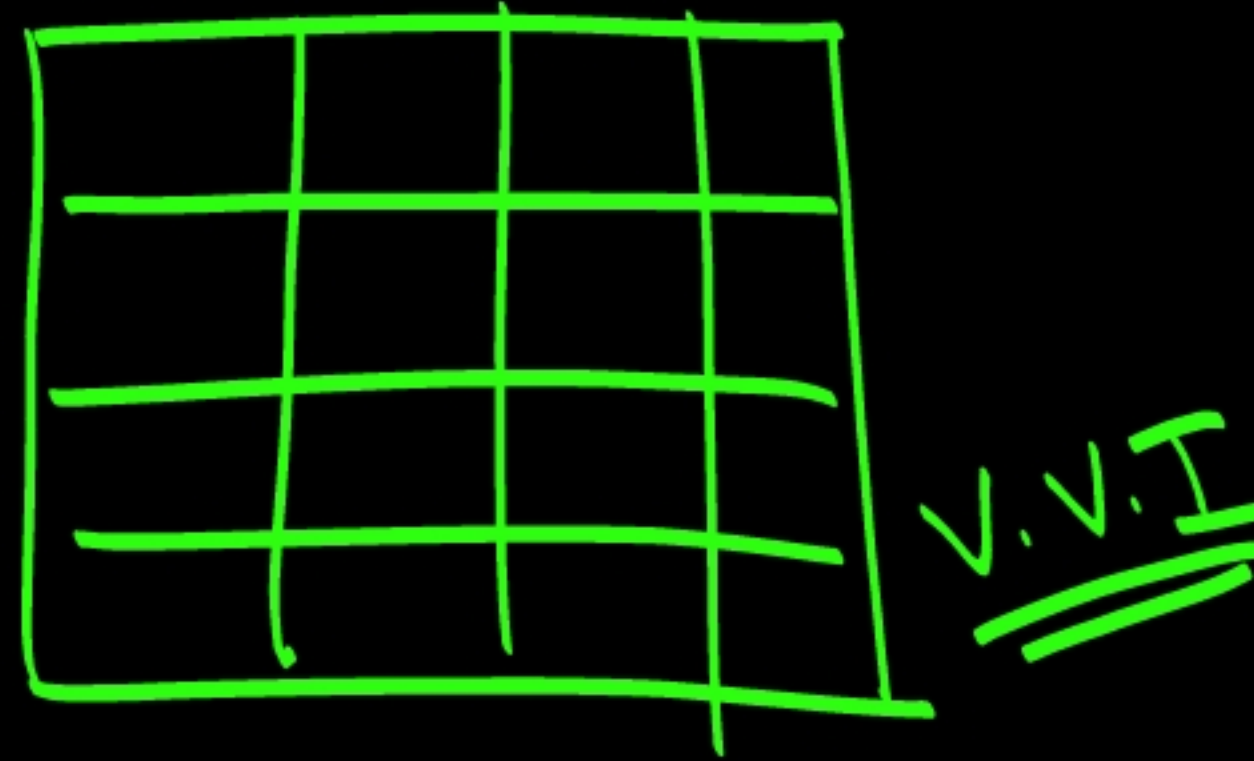
(D) Queue

(32) A Threaded Binary Tree makes use of:

- (A) Multiple Roots**
- (B) NULL pointers of leaf nodes ✓**
- (C) Extra stacks**
- (D) Double pointers**

(33) Priority Queue can be efficiently implemented using:

- (A) Array**
- (B) Linked List**
- (C) ☒ Heap**
- (D) Stack**



(34) The time required to find the shortest path between all pairs of nodes in a graph is best given by:

- (A) Dijkstra's**
-  **(B) Floyd-Warshall Algorithm**
- (C) Kruskal's**
- (D) Bellman-Ford**

THANK YOU